# CO2 ASSESSMENT TOOL 3 DEBUGGING & OPTIMISATION TASK

## Aim

Debug brute force Convex Hull logic for collinear cases.

## Bug Explanation

The original code rejects every collinear point (orientation==0). Hull edges may legitimately contain collinear boundary points. Only points lying outside the candidate edge should invalidate it.

## Corrected Algorithm

Determine orientation for all points. Accept collinear points. Reject only if points exist on both sides. For collinear points ensure they lie on the segment.

## Improved Codelike Python

```
def orientation(a,b,c):
  return (b[0]-a[0])*(c[1]-a[1])-(b[1]-a[1])*(c[0]-a[0])

def on_segment(a,b,p):
  return min(a[0],b[0])<=p[0]<=max(a[0],b[0]) and min(a[1],b[1])<=p[1]<=max(a[1],b[1])

def is_hull_edge(p1,p2,points):
  side=0
  for p in points:
    if p==p1 or p==p2:
      continue
    val=orientation(p1,p2,p)
    if val==0:
      if not on_segment(p1,p2,p):
        return False
      continue
    cur=1 if val>0 else -1
    if side==0:
      side=cur
    elif side!=cur:
      return False
  return True
```

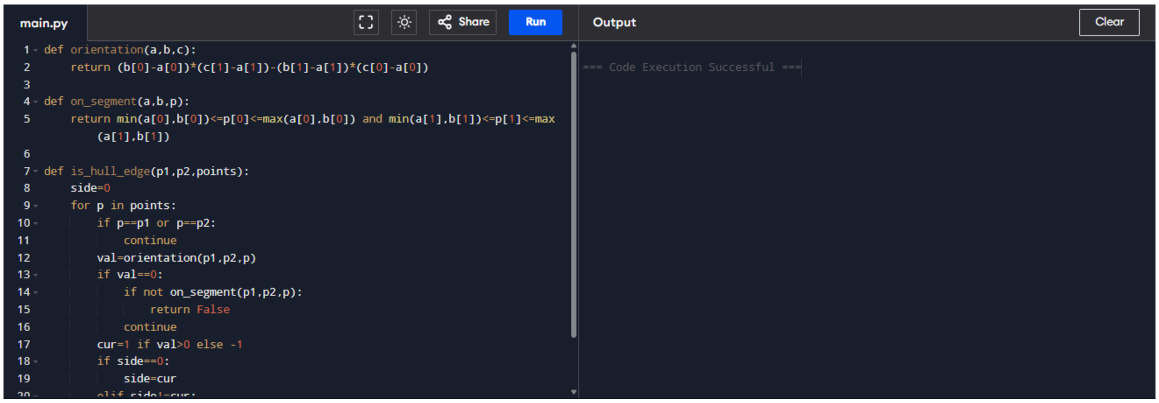
## Optimization

Only one orientation computation per point. Side tracking avoids repeated comparisons.

## Time Complexity

Brute Force: O(n^3). Orientation checks are O(1).

## Output Screenshot




## Conclusion

Correct handling of collinear boundary points ensures valid hull edges while preserving brute-force complexity.